Project Title

Daytime Population Trends in the Bay Area
Before and After the Covid Pandemic

Group Number

18

Group Name

YAGviz

Team Members

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Idea 4: Daytime Population

GitHub Repository

https://github.com/CP101-FA26-Group-18/Final-Project

References

Spear, B. (2011) Improving employment data for transportation planning (Technical Report NCHRP 08-36 Task 098). Cambridge, MA: Cambridge Systematics. https://onlinepubs.trb.org/onlinepubs/nchrp/docs/NCHRP08-36(98)_FR.pdf
(Supports our decision to use census tracts as the finest spatial granularity level)

Research Question

Our original research question is: How did the nine-county Bay Area's daytime population patterns change at the county and census-tract levels from 2017 to 2023, and how did these patterns differ before and after the COVID-19 pandemic? Our analysis focuses on employment-reported home-to-work relationships in which both the workplace and residence are located within the nine Bay Area counties.

Based on feedback provided by Professor Hosseini and our MTC mentor Kaya, we have modified our approach since our first proposal. Initially, we intended to adjust LODES commuter flows at the census-tract level using ACS work-from-home (WFH) rates. However, yearly ACS surveys do not provide reliable WFH rates at the tract level and tract level 5 year estimates cannot be reliably applied to yearly LODES data. Instead of directly modifying tract level flows, we choose to use LODES as our primary dataset and ACS county-level WFH rates as supporting data. An additional change, as suggested by Hosseini, is the inclusion of a third level of geospatial granularity, the city level. Census TIGER shapefiles were incorporated into the project to facilitate grouping tracts into cities cleanly.

In summary, our current analysis focuses on changes in employment-reported commuting patterns as a proxy for daytime population trends rather than directly measuring the number of people physically present during the workday.

Data Cleaning and Spatial Processing

Our primary dataset is the US Census Bureau's LODES8 (2017–2023), using California-specific all-job files that exclude out-of-state commuters and filtered for OD pairs where both home and work blocks are within the nine counties. This dropped about 80% of statewide records (e.g., for 2023 13.27 million entries were dropped, leaving 2.97 million). We then aggregated these block pairs to tract pairs based on 11-digit FIPS codes, which further compressed each yearly file by about 74% (e.g., 767,054 entries in 2023) while conserving all commuter volume. Aggregating to the county level was done similarly, with 6-digit FIPS codes. We chose census tracts as our finest resolution to avoid the noise and map overplotting inherent to block-level data, as tract level aggregates are a reliable proxy (Spear, 2011). Also, running 7 years of block-level files in Python triggered memory crashes in our environment (VSCode), making the tract level data unfeasible to work with. A single tract-level dataset is about 25% of the block-level dataset in size, yet requires roughly 1 GB of RAM.

For city spatial boundaries, we took the TIGER shapefile dataset, dropped 7,356 non-Bay tracts (80.6%), and reprojected the remaining 1,772 Bay Area tracts and municipal boundaries to EPSG 2227. Because census tracts and municipal boundaries do not naturally align, we applied the largest-share rule using gpd.overlay(how="intersection") to assign each tract to the city or Census Designated Place (CDP) covering the largest share of its area. This produced a clean one-to-one mapping of 1,767 tracts across 179 municipalities, with the remaining 5 tracts categorized as "Unincorporated" to prevent artificial population inflation in adjacent cities. This way we were able to group annual tract OD pairs into roughly 22,000 city pairs. As a sanity

check, we confirmed that total commuter counts matched across tract, city, and county aggregations (e.g., exactly 3,288,681 commuters in 2023).

To supplement commute flows with WFH data, we used county-level ACS 1-year estimates from Table B08301 for 2017–2019 and 2021–2023 (2020 is not provided due to survey collection disruptions during the pandemic). We cleaned each file by removing the descriptive header row, dropping all 21 margin-of-error columns, slicing geographic IDs to 5-digit county FIPS, and calculating WFH rates as remote workers (B08301_021E) divided by total workers (B08301_001E). Because Napa County lacked 1-year estimates in 2021 and 2022 (due to small sample sizes), we assigned those entries as NaN rather than interpolating values to avoid skewing the data. Keeping ACS WFH rates strictly at the county level avoids the problems of 5-year ACS data while still allowing us to perform some regional analysis on WFH rates.

More thorough explanation and logs of the data cleaning process can be found in the markdown and text files in the data subfolder of the project directory.

Methodology

Our workflow begins with the cleaning and filtering of the annual LODES OD datasets to keep only the home-to-work connections within the nine-county Bay Area. Then, we aggregate Census-block OD records into census tracts, counties, and cities.

For our county-level visualization, we use gpd.sjoin( ) to match Census tract representative points with county boundaries. Before the spatial join, the geographic datasets are projected to EPSG:2227 (NAD83/California State Plane Zone 3). Next, we compute the inbound, outbound, and net LODES job flows using groupby( ) and sum( ), and we utilize Matplotlib and GeoPandas to show the results. Produced charts and visualizations include the following:

- A line chart of the count of all LODES origin-destination pairs within the Bay Area, with year on the x-axis (2017–2023) and the count on the y-axis.
- A grouped bar chart of the count of inbound and outbound LODES origin-destination data, with categories on the x-axis representing counties and the count on the y-axis.
- A bar chart of the percent change in LODES origin-destination data between 2019 and 2023, by the 9 Bay Area counties.
- A grouped bar chart of county WFH rates from ACS 1-year estimates, in 2019 and 2023.
- A bar chart of the count of LODES origin-destination jobs in the 10 cities with the highest count of destination points in 2023.
- A map of the 9 Bay Area counties colored by the count of the net-flow of LODES origin destination data.

Our validation confirms that the 3,288,681 jobs in 2023 were retained during the spatial join and that all 1,772 Bay Area tracts were matched.

Early Results

According to our initial estimates, the total number of LODES jobs in the Bay Area increased from roughly 3.27 million in 2017 to 3.39 million in 2019 before falling to 3.07

million in 2021 (a 9.5% decline from 2019). Although the number increased to 3.29 million by 2023, it was still roughly 3.0% below the pre-pandemic level. This is roughly consistent with published economic reports on the Bay Area.

According to our county-level map from 2023, San Francisco had the biggest net inflow of jobs from other Bay Area counties (roughly 205,000), while Contra Costa had the biggest net outflow (around 143,000). It's interesting to note that, despite an 8.7% drop in workplace-based LODES jobs between 2019 and 2023, San Francisco continued to be a top employment destination. Our ACS comparison also shows that San Francisco's WFH rate rose from roughly 7.4% in 2019 to 24.4% in 2023.

These results suggest changes in regional employment and commuting patterns before and after COVID-19. However, these data shouldn't be considered as real daily commuter counts or daytime population because LODES shows administrative home-to-work relationships rather than observed daily trips. Further analysis combining WFH data and LODES data is necessary to estimate real commuter counts.

Plan to Finish

Nov. 6–15 (Goryune): Finalize the data-cleaning workflow, verify the consistency of aggregated LODES data, and complete the documentation needed for reproducibility.

Nov. 6–15 (Aaron): Review alternative WFH data sources recommended by Hosseini and evaluate whether they can reasonably supplement our LODES analysis. Specifically, remote-adjusted tract level data (mean imputation, spatial disaggregation).

Nov. 16–Dec. 1 (Yaqi): Refine the visualizations, develop at least two interactive or animated HTML maps in Python, and examine notable spatial changes at tract and county levels.

Dec. 2–14 (All members): Complete validation and interpretation, finalize the GitHub Pages website and repository documentation, and prepare the final report and presentation.

Challenges

Our main challenges are the memory requirements for processing large LODES datasets and the lack of robust tract-level WFH data. For the former, we will use cleaned CSV files for producing visualization in one file while maintaining a separate file for reproducible data cleaning code. We shall clearly point out this restriction, and our data cleaning file still meets the "runs top-to-bottom cleanly" project requirement outlined on the course website. For the latter, we will concentrate on analyzing LODES home-to-work relationships (e.g. explore patterns in demographics, workplace types) if we are unable to find an appropriate WFH adjustment. We would appreciate feedback on whether using county-level ACS WFH rates as supporting data is sufficient, as our research into alternative WFH datasets has not been fruitful so far.

A third challenge is in finding interesting and engaging ways to explore patterns in the data. We hope to point out smaller scale patterns found within the broader regional set, as suggested by Hosseini. We would appreciate feedback regarding the methodology we should use to determine which temporal and spatial patterns to explore further for our final visualizations.